

Pham Minh Dang

BLOCKCHAIN ENGINEER

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‘You are always in somebody school’”

Summary

Blockchain Engineer with hands-on experience in smart contract development, cross-chain bridges, DeFi protocols, and blockchain infrastructure.

Alongside industry work, I have been pursuing research in blockchain scalability and data availability. My first research paper, Efficient Data Availability Sampling via Coded Distributed Arrays, was accepted to IEEE ICBC 2026.

I am currently interested in consensus, data availability, consensus, and scalable decentralized systems.

Work Experience

Oraichain Lab

Hanoi, Vietnam

SMART CONTRACT ENGINEER

Feb. 2024 – July. 2025

- Core developer of the OraibTC bridge (Bitcoin–Oraichain), involved in both phases: the initial subnet (Bitcoin light-client verification, application-hash debugging, LCD module) and the second-generation CosmWasm mainnet deployment (smart contracts, off-chain relayers, frontend integration). The bridge reached approximately \$800K trading volume in its first two weeks.
- Core developer of a cross-chain asset bridge on the TON blockchain, implemented in Func with IBC-inspired design. Worked on smart contract development, cross-chain security, and consensus flow review. The bridge reached a total value locked of about \$200K across both chains.
- Core developer of a message relay system between Solana and Oraichain for a centralized bridge prototype, designed for high-throughput relaying and transaction integrity in asynchronous environments. The bridge reached over \$2M in total value locked at its peak with a high volume of daily transactions.
- Designed and built the Perpetual Whales MCP to aggregate and profile historical on-chain whale trading behavior and portfolio statistics. Developed data crawlers across multiple sources, providing a reference dataset for perpetual trading insights in the Thesis platform, with outputs integrated into an LLM for analytics-driven trading reports.
- Participated in the development of REWOO agents within an AI agentic platform, integrating Google’s A2A protocol for inter-agent communication. Implemented parallel MCP tool calls to reduce token consumption compared to Codeact agents.
- Contributed code to Microsoft’s open-source Autogen framework, adding compatibility for MCP integration and ensuring correct result handling with the Magnetic One agent.

Orbiton Labs

Hanoi, Vietnam

CO-FOUNDER · BUILDER

Oct. 2024 – July. 2025

- Architected a concentrated liquidity AMM (CLAMM) protocol inspired by Uniswap V3, tailored for deployment on the TON blockchain with batch tick model to align with TON’s computation architecture and state layout
- Developed core smart contracts in Func, implementing logic for liquidity provision, tick management, and swap path calculation.
- Built an off-chain indexer to synchronize on-chain events and maintain real-time market state, enabling analytics and frontend integration.

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Hanoi, Vietnam

BLOCKCHAIN DEVELOPER

Sep. 2022 – Nov. 2023

- Core developer of an NFT marketplace platform (team size: 8), responsible for EVM-compatible smart contracts and overall blockchain architecture.
- Contributed to backend services, building an indexer system to synchronize on-chain smart contract data, and supported the frontend team in integrating Web3 functionalities with ethers.js and wallet-adapter.
- Mentored Web2 developers and interns, delivering training on blockchain fundamentals, wallet integration, and smart contract interaction.

Pippip - Laboratory in University

Hanoi, Vietnam

FRONTEND DEVELOPER

Mar. 2021 – Apr. 2022

- Developed and maintained customer- and admin-facing web interfaces for a long-distance car booking platform, focusing on responsiveness and usability.
- Collaborated with backend developers to define APIs based on product requirements and integrated real-time booking updates via WebSocket.
- Assisted onboarding of new frontend interns by providing project guidance and support.

Technical Stack

Back-end	Javascript, Typescript, Rust, Express.js, NestJS, Golang
Smart Contracts	Solidity, Rust, FunC, Move
Front-end	TypeScript, React, Next.js, TailwindCSS
DevOps	AWS Services, GitHub Actions

Honors & Awards

INTERNATIONAL

2025	2nd Place , Scoop AI Hackathon	Online
2024	1st Place (DeFi Track) & 3rd Place (DeDust Track) , TON Hackers League Hackathon — Winter Edition	Online

DOMESTIC

2024	Best Presentation Award , Reviewer Report for Engineering Graduation Thesis — Hanoi University of Science and Technology	Hanoi, Vietnam
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Writing

Galois Gossip: When Algebra Meets Ethereum's Gossip Layer

Medium Page

BLOGGER

Nov. 2025

- Explain how Galois Gossip of Optimum team works which is described in the paper "OPTIMUMP2P: Fast and Reliable Gossiping in P2P Networks", which outperforms the Gossip Protocol of Ethereum

Ethereum Consensus Layer: From Peer Discovery with Discv5 to Message Propagation with Gossipsub

Medium Page

BLOGGER

Sep. 2025

- Deep dive on ethereum consensus layer from distributed hash table to gossip protocol

Layer 2 Explained: From Ethereum State Transitions to Rollups

Medium Page

BLOGGER

Aug. 2025

- Explain how L2 interacts with L1 through Ethereum State Transition

Education

Hanoi University of Science and Technology (HUST)

Hanoi, Vietnam

ENGINEER'S DEGREE IN COMPUTER SCIENCE AND ENGINEERING (WITH HONORS)

Sep. 2019 – Sep. 2024

- Graduated with Honors, GPA: 3.62/4.0

Presentation

Blockchain Course, BKC Fintech

Remote

PRESENTER — "Building a Mini Bitcoin with JavaScript"

2024

- Developed and conducted a 2-week hands-on workshop introducing Bitcoin fundamentals through code-based instruction.
- Led participants in building a simplified Bitcoin-like blockchain in JavaScript, covering transaction validation, mempool simulation, and Proof-of-Work consensus.
- Bridged theory and implementation to help learners gain a foundational understanding of decentralized ledger design.

Online Course via Udemy

Remote

INSTRUCTOR — "Real-time NFT Auction with Solidity - Blockchain Dapp"

2023

- Designed and delivered an online course introducing smart contract development through a real-world use case: a decentralized NFT auction platform.
- Guided learners in implementing Solidity-based auction logic and integrating Moralis to enable real-time event handling and off-chain data sync.
- Taught technical concepts including contract-to-frontend communication, event listeners, and NFT metadata management to over 30 enrolled students.

Extracurricular Activity

SINNO Club — Hanoi University of Science and Technology (HUST)

Hanoi, Vietnam

MEMBER & DEPUTY OF LOGISTICS DIVISION

2020 – 2022

- Managed and maintained the internal server infrastructure used by SINNO members for deploying and testing club projects.
- Supported student-led product teams by providing a shared development environment and infrastructure troubleshooting.